SSL Manager — User Guide

SSL Manager — User Guide

Table of Contents

1. Connecting to SSL Manager
2. Logging In
3. Dashboard Overview
4. Certificate Profiles
5. Certificate Chains
6. Certificate Authorities (Internal CA)
7. Certificate Lifecycle
8. Browsing and Finding Certificates
9. Downloading Certificates
10. Renewing a Certificate
11. User Management
12. Audit Log
13. Expiry Notifications

SSL Manager — User Guide

SSL Manager is a web application for managing the full lifecycle of SSL/TLS certificates — from generating a private key and Certificate Signing Request (CSR) through to storing the signed certificate and downloading it in the format required by your server.

This guide covers everything an end user needs: connecting to the application, setting up your environment, and managing certificates day-to-day.

Table of Contents

1. Connecting to SSL Manager
    - macOS and Linux
    - Windows — OpenSSH (Windows 10/11)
    - Windows — PuTTY
    - Persistent SSH Config (macOS / Linux)
    - Keeping the Tunnel Open
2. Logging In
3. Dashboard Overview
4. Certificate Profiles
    - Create a Profile
    - Set the Default Profile
    - Edit or Delete a Profile
5. Certificate Chains
    - Create a Chain
    - Add Intermediate Certificates
    - Import a Bundle
6. Certificate Authorities (Internal CA)
    - Create a CA
    - Sign a Pending Certificate
    - Download the CA Certificate
    - Delete a CA
7. Certificate Lifecycle
    - Create a New Certificate
    - Download and Submit the CSR
    - Upload the Signed Certificate
    - Import an External CSR
    - Import a PKCS#12 / PFX Certificate
    - Import a Private Key + Certificate
8. Browsing and Finding Certificates
9. Downloading Certificates
10. Renewing a Certificate
11. User Management
12. Audit Log
13. Expiry Notifications

1. Connecting to SSL Manager

SSL Manager is designed to be accessible only from the server it runs on. To reach it from your computer you use **SSH port forwarding**, which creates an encrypted tunnel through your existing SSH connection. No extra firewall ports need to be opened on the server.

> **What this means in practice:** you open a terminal, run one SSH command, then open your browser to `http://localhost:5001`. The browser talks to the server securely through the SSH tunnel.

1.1 macOS and Linux

Open a terminal and run:

```
ssh -L 5001:127.0.0.1:5001 youruser@your-server
```

Replace `youruser` with your server username and `your-server` with the server's hostname or IP address. If the server uses a non-standard SSH port, add `-p PORT`:

```
ssh -L 5001:127.0.0.1:5001 -p 2222 youruser@your-server
```

While this terminal session remains open, browse to:

```
http://localhost:5001
```

1.2 Windows — OpenSSH (Windows 10/11)

Windows 10 and 11 include OpenSSH. Open **Command Prompt** or **PowerShell** and run:

```
ssh -L 5001:127.0.0.1:5001 youruser@your-server
```

Then open your browser to `http://localhost:5001`.

Persistent SSH config on Windows 10 / 11

Windows 10 (build 1809 and later) and Windows 11 support the same `~/.ssh/config` file as macOS and Linux. The file lives at:

```
C:\Users\YourUsername\.ssh\config
```

Create or edit it in any text editor (Notepad, VS Code, etc.) using exactly the same syntax:

```
Host ssl-manager
    HostName     your-server
    User         youruser
    LocalForward 5001 127.0.0.1:5001
    ServerAliveInterval 60
    ServerAliveCountMax 10
```

After saving, connect from PowerShell or Command Prompt with:

```
ssh ssl-manager
```

If the `.ssh` folder does not exist yet, run `ssh youruser@your-server` once from PowerShell — Windows creates the folder automatically on first use.

1.3 Windows — PuTTY

1. Open PuTTY and enter your server's hostname under **Session**.
2. In the left panel navigate to **Connection** → **SSH** → **Tunnels**.
3. Fill in:
    - **Source port:** `5001`
    - **Destination:** `127.0.0.1:5001`
    - Select **Local**
4. Click **Add**. You will see `L5001  127.0.0.1:5001` in the forwarded ports list.
5. Go back to **Session**, save the session if you like, then click **Open**.

While the PuTTY session is connected, browse to `http://localhost:5001`.

1.4 Persistent SSH Config (macOS / Linux)

To avoid typing the full command every time, add an entry to `~/.ssh/config`:

```
Host ssl-manager
    HostName     your-server
    User         youruser
    LocalForward 5001 127.0.0.1:5001
```

After saving the file you can connect with just:

```
ssh ssl-manager
```

Then open `http://localhost:5001`. The `Host` label (`ssl-manager`) can be anything you like.

1.5 Keeping the Tunnel Open

The tunnel is active for as long as the SSH session is open. If you close the terminal or the session times out, refresh the SSH connection and reload the browser tab.

To keep a session alive longer, add to `~/.ssh/config`:

```
Host ssl-manager
    ...
    ServerAliveInterval 60
    ServerAliveCountMax 10
```

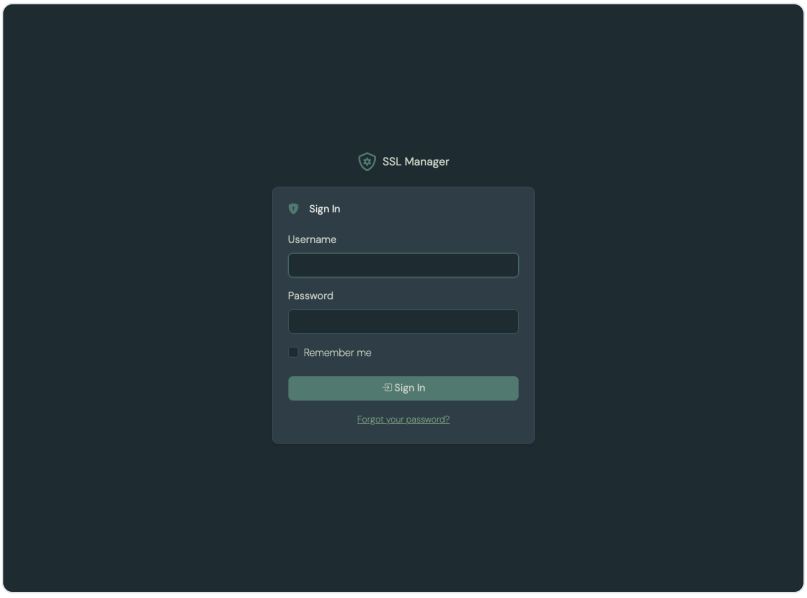
This sends a keepalive packet every 60 seconds, keeping idle sessions open for up to 10 minutes of inactivity.

---

2. Logging In

Navigate to `http://localhost:5001` in your browser while the SSH tunnel is active.

- **First visit ever:** you are redirected to the **Setup** page to create the initial administrator account. Fill in a username, email, and password and click **Create Admin Account**. You are logged in automatically.
- **All subsequent visits:** you are presented with the login form. Enter your username and password.



Login page

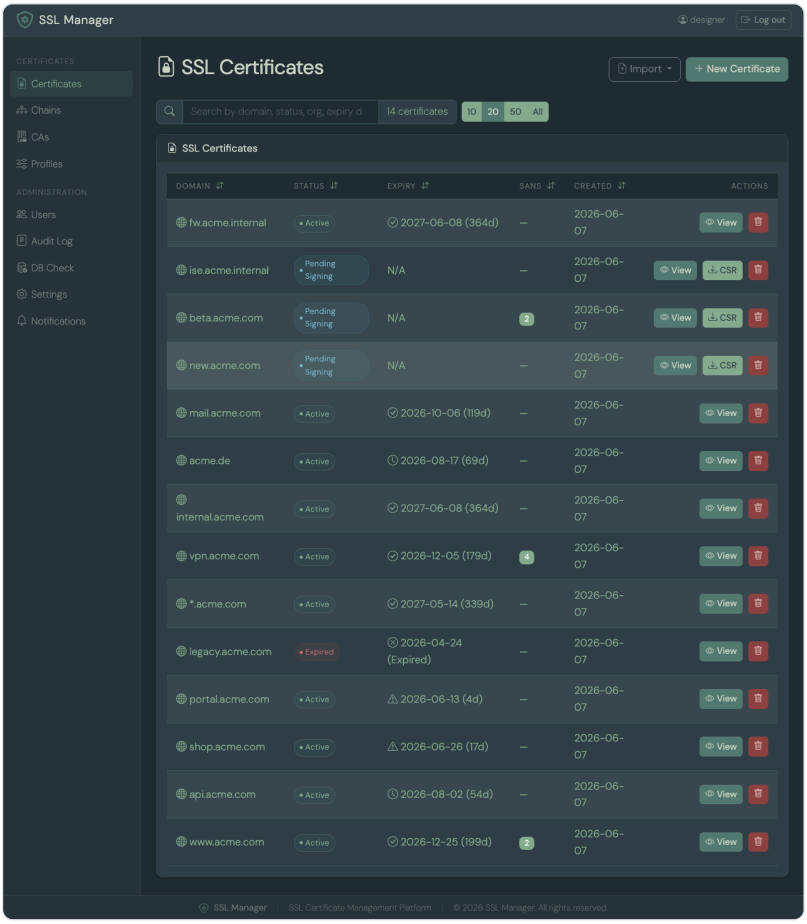
If your session expires you are returned to the login page automatically.

---

3. Dashboard Overview

After logging in you land on the **Certificates** page. The navbar at the top provides access to all sections:

Nav item	Purpose
SSL Manager (logo)	Return to the certificates list from anywhere
Certificates	List, search, sort, and manage all certificates
Chains	Manage named sets of intermediate CA certificates
CAs	Manage internal Certificate Authorities and sign pending certificates
Profiles	Manage certificate subject templates (org name, country, key size, etc.)
Users	Add, edit, or remove users *(superadmin only)*
Audit Log	View a history of all actions taken in the app *(superadmin only)*
Logout	End your session



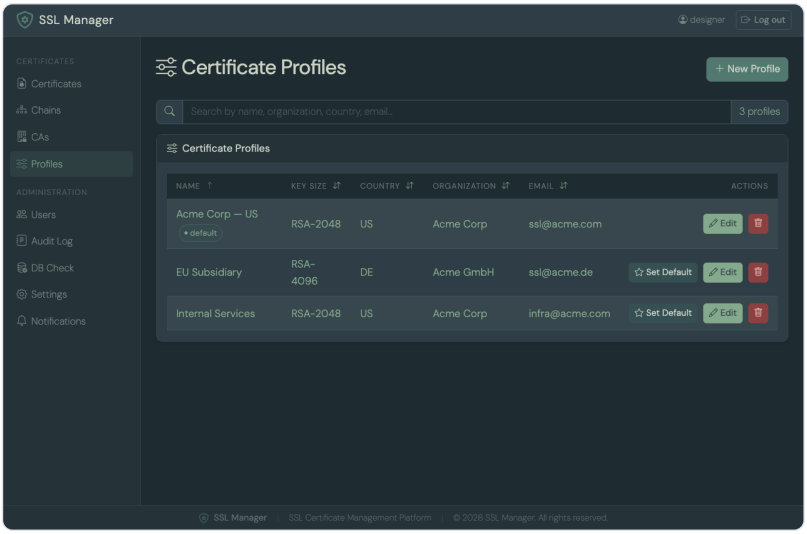
Certificates list — dashboard overview

---

4. Certificate Profiles

A **profile** is a reusable template for the subject fields that appear in every CSR — your organisation name, country, key size, and so on. Instead of filling in these fields from scratch each time you create a certificate, you pick a profile and the form is pre-filled for you.

You can have as many profiles as you need. For example: - `Acme Corp — US` for US-based certificates - `Acme EU GmbH` for European subsidiary certificates - `Internal Services` with a different org unit



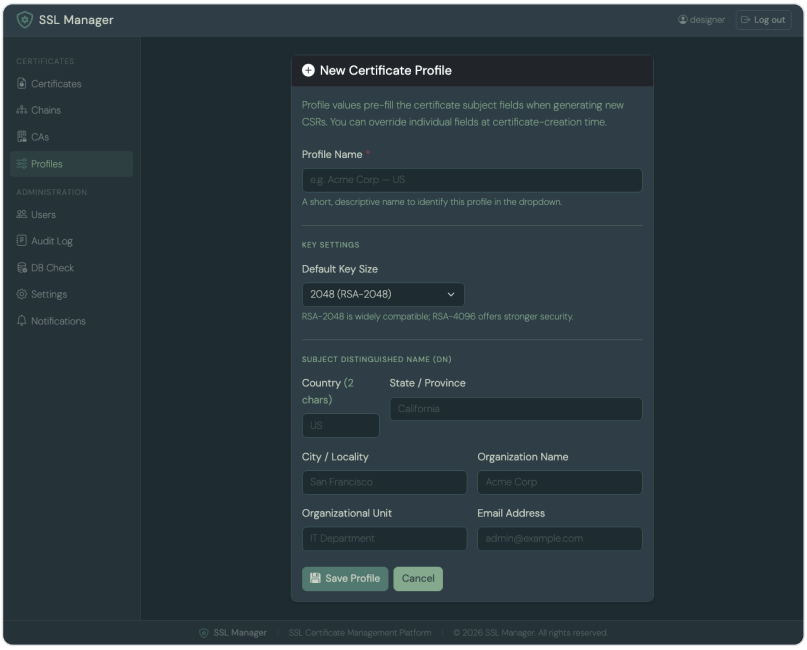
Profiles list

4.1 Create a Profile

1. Navigate to **Profiles** in the navbar.
2. Click **New Profile**.
3. Fill in:

Field	Description	Example
Profile Name	A short descriptive label for this profile	`Acme Corp — US`
Key Size	RSA key length: `2048` (standard) or `4096` (stronger)	`2048`
Country	Two-letter ISO country code	`US`
State / Province	State or province	`California`
City / Locality	City	`San Francisco`
Organization Name	Your organisation's legal name	`Acme Corp`
Organizational Unit	Department or team	`IT`
Email Address	Contact email embedded in the CSR	`ssl@acme.com`

1. Click **Save Profile**.



New profile form

4.2 Set the Default Profile

The profile marked **default** (green badge) is used automatically when only one profile exists, or pre-selected when creating a new certificate with multiple profiles.

To change the default:

1. Go to **Profiles**.
2. Click **Set Default** next to the profile you want to promote.

Only one profile can be the default at a time. The previous default is unset automatically.

4.3 Edit or Delete a Profile

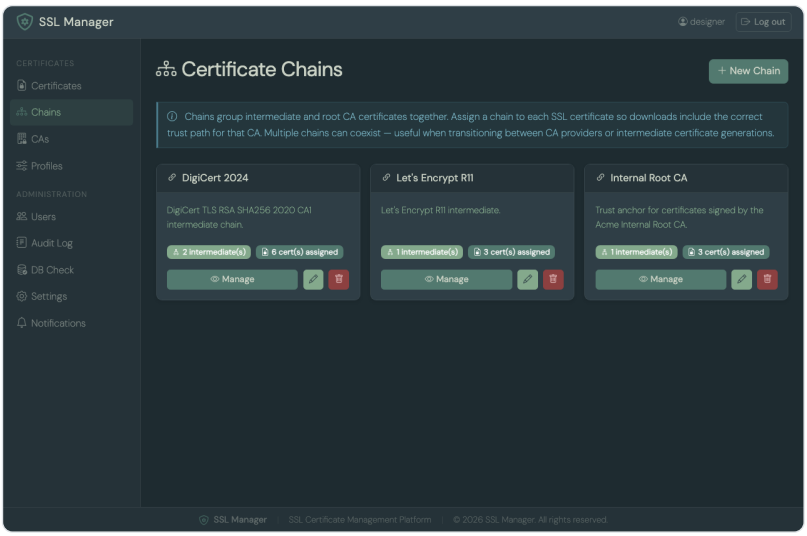
- **Edit:** click the **Edit** button next to a profile, update the fields, and click **Save Profile**.
- **Delete:** click **Delete** and confirm. You cannot delete the last remaining profile.

> Changing or deleting a profile does not affect certificates that were already created — each certificate stores a permanent copy of the subject fields at the time it was generated.

5. Certificate Chains

A **chain** is a named collection of the intermediate (and optionally root) CA certificates that connect your domain certificate back to a trusted root. You must set up at least one chain before you can generate full bundled download formats.

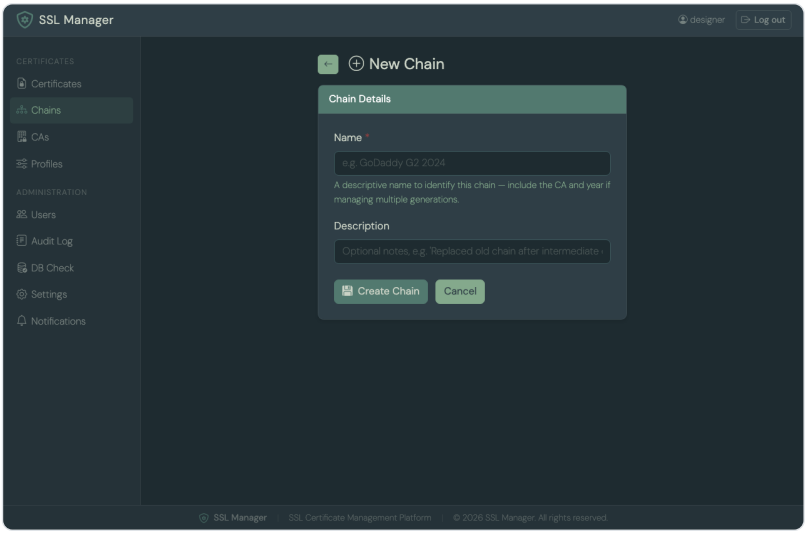
Common scenarios for multiple chains: - You use different Certificate Authorities for different certificate types. - Your CA rotated their intermediate certificate — you need the old chain for existing certs and the new chain for new ones.



Chains list

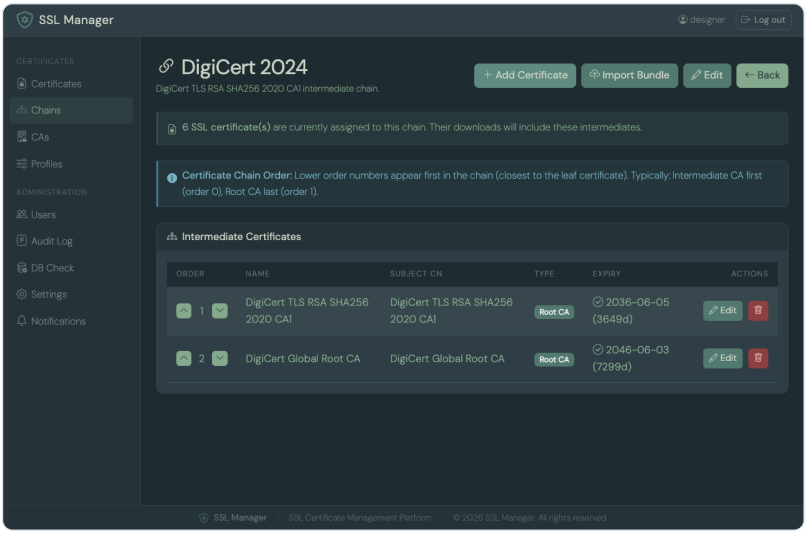
5.1 Create a Chain

1. Navigate to **Chains** in the navbar.
2. Click **New Chain**.
3. Enter a **Name** (e.g. `DigiCert 2024`) and an optional **Description**.
4. Click **Save**.



New chain form

You are taken to the **Chain Detail** page where you add the individual CA certificates.



Chain detail page

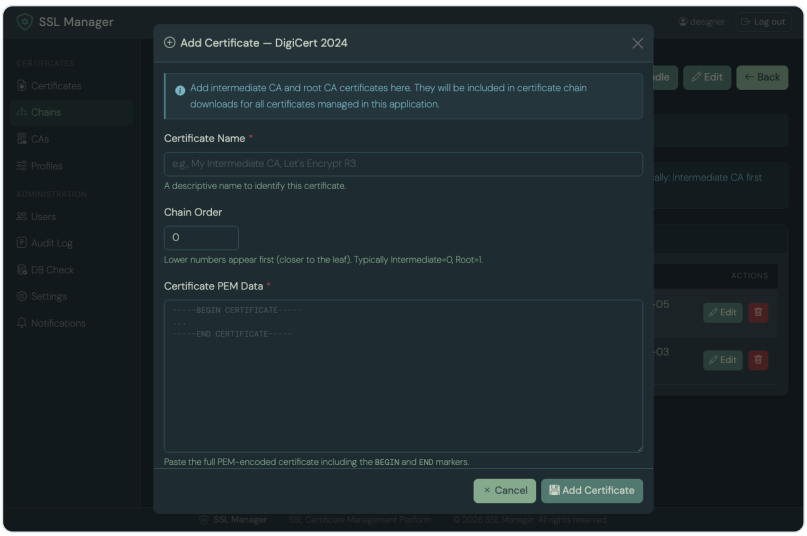
5.2 Add Intermediate Certificates

From the **Chain Detail** page, click **Add Certificate**. Fill in:

Field	Description
Display Name	A label for this entry, e.g. `DigiCert TLS RSA SHA256 2020 CA1`
Order	Position in the chain. `1` = the intermediate that directly signed your domain cert; higher numbers go toward the root

Field	Description
PEM Data	The certificate in PEM format, including the `-----BEGIN CERTIFICATE-----` and `-----END CERTIFICATE-----` lines

Click **Save**. Repeat for each certificate in the chain (typically an intermediate and a root).



Add certificate form

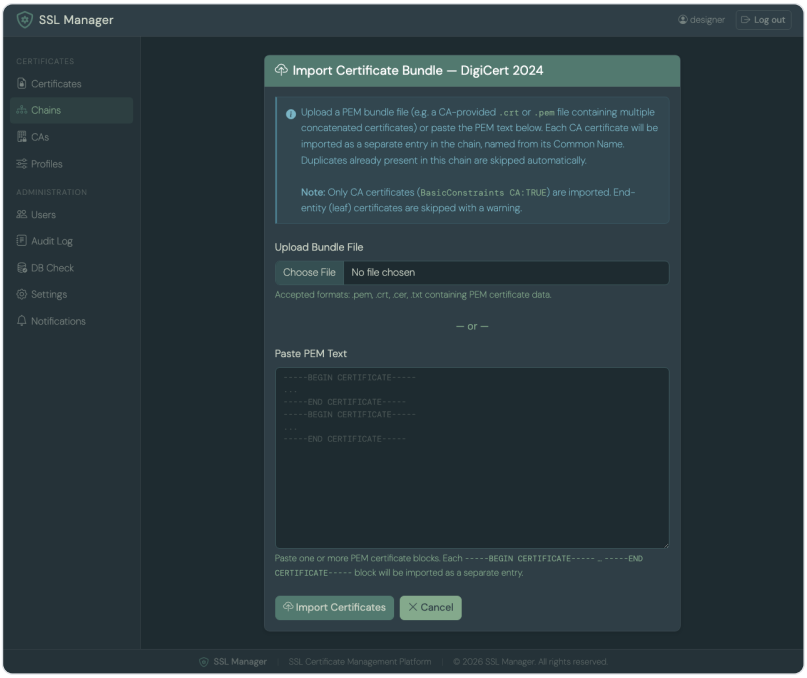
Use the ▲ / ▼ chevron buttons on each row to reorder entries on the Chain Detail page.

5.3 Import a Bundle

If your CA provides a single file containing multiple certificates concatenated together (common with `.crt`, `.pem`, or `.ca-bundle` files), use **Import Bundle** instead of adding them one by one:

1. On the **Chain Detail** page, click **Import Bundle**.
2. Paste the full PEM contents or upload the file — the app splits it into individual certificates automatically.
3. Review the detected certificates and click **Import**.

Each certificate is added as a separate entry in the chain, ordered from the first PEM block to the last.



Import bundle page

> **End-entity certificates are skipped:** Chains should only contain CA certificates (intermediates and roots). If the bundle includes a domain/leaf certificate, SSL Manager detects it automatically and skips it with a warning. Only CA certificates are imported into the chain.

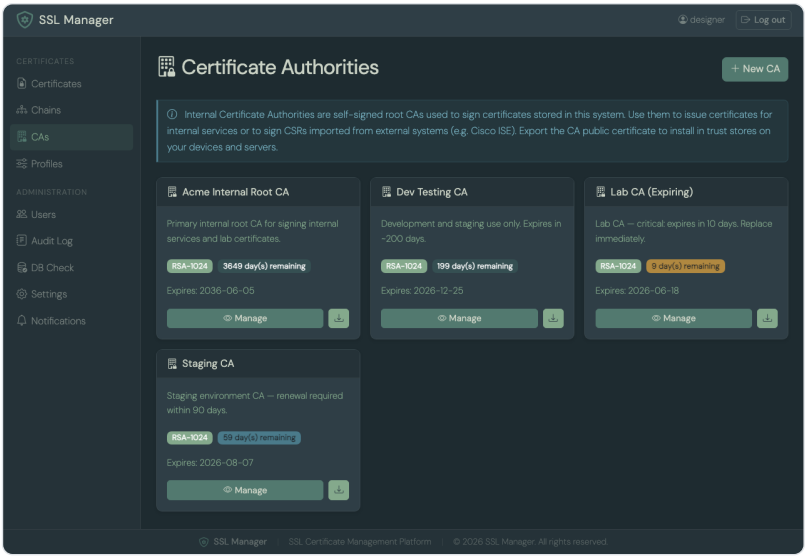
---

6. Certificate Authorities (Internal CA)

SSL Manager includes a built-in Certificate Authority module. An internal CA lets you sign certificates entirely within SSL Manager — no external CA portal required. This is useful for:

- Internal services and development environments that don't require a publicly-trusted certificate.
- Network equipment such as Cisco ISE, firewalls, and load balancers that generate their own CSRs and need a signed certificate returned to them.
- Lab and staging environments where you control the full trust chain.

Each CA is a self-signed root CA. You can create multiple CAs — for example one per environment (dev, staging, prod) or one per team.



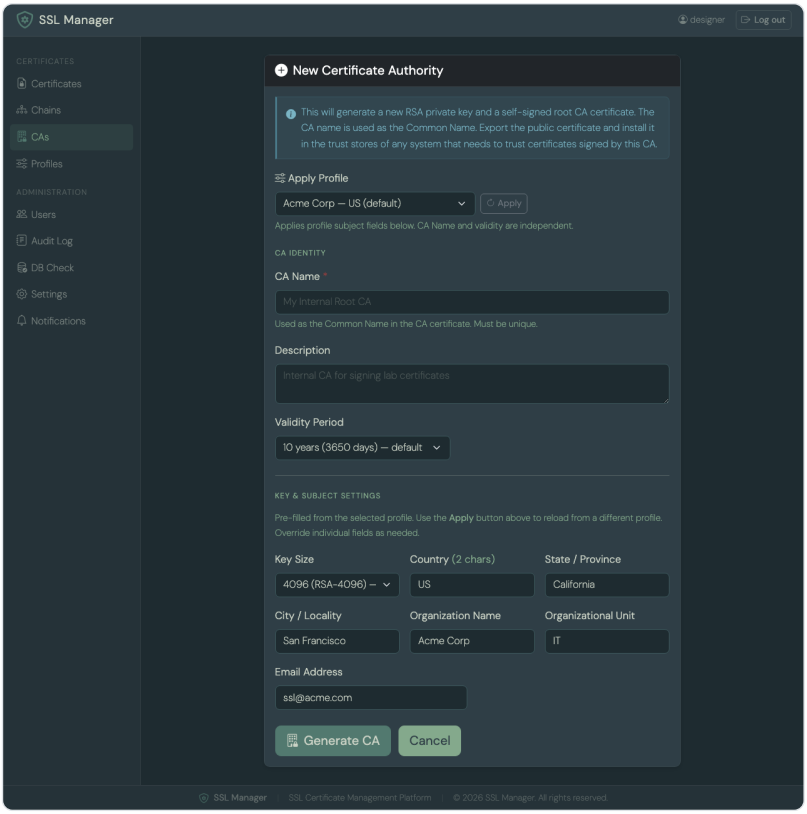
CAs list

6.1 Create a CA

1. Navigate to **CAs** in the navbar.
2. Click **New CA**.
3. Fill in the form:

Field	Description	Default
CA Name	A short descriptive label, also used as the Common Name	—
Description	Optional free-text note	—
Validity Period	How long the CA certificate remains valid	10 years
Key Size	RSA key length: `2048` or `4096`	4096
Country	Two-letter ISO country code	From profile
State / Province	State or province	From profile
City / Locality	City	From profile
Organization Name	Organisation's legal name	From profile
Organizational Unit	Department or team	From profile
Email Address	Contact email embedded in the CA certificate	From profile

1. Click **Create CA**.



New CA form

The application generates an RSA private key (which never leaves the server) and a self-signed CA certificate. You are taken to the **CA Detail** page.

> **Key security note:** The CA private key is stored on the server alongside certificate private keys. Treat it with the same sensitivity as any other private key material. See the backup note in the installation guide for recommendations on securing backup archives.

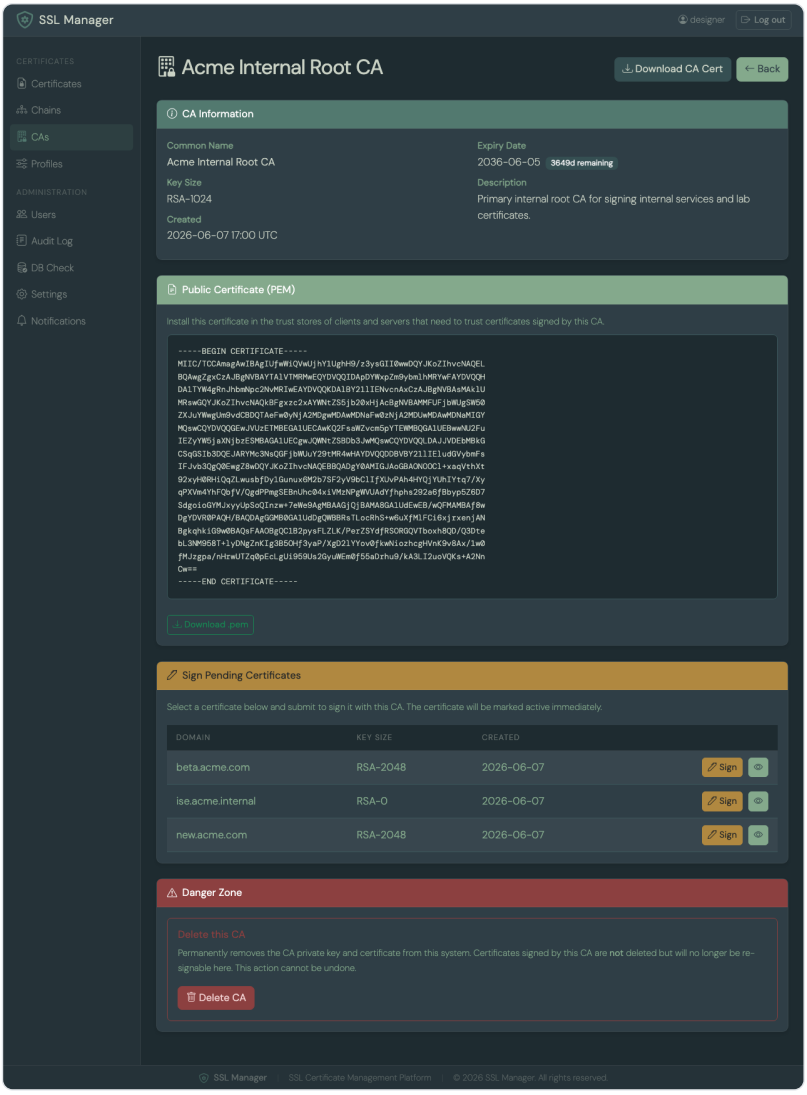
6.2 Sign a Pending Certificate

Any certificate with status **Pending Signing** can be signed by an internal CA — whether it was created using **New Certificate** or imported using **Import CSR**.

From the CA Detail page:

1. Navigate to **CAs** and click the CA name.
2. The **Pending Certificates** table lists all certificates awaiting a signature.
3. For each certificate, set the **Validity (days)** field (default: 365).
4. Click **Sign** for the certificate you want to sign.

The certificate status changes to **Active** and the expiry date is set automatically.



CA detail — Sign Pending Certificates

From the Certificate Detail page:

If you are already viewing a pending certificate, a **Sign with Internal CA** card appears on the Certificate Detail page:

1. Select the CA to use from the dropdown.
2. Set the validity period in days.
3. Click **Sign Certificate**.

6.3 Download the CA Certificate

To configure a device or browser to trust certificates signed by your internal CA, you need to install the CA certificate as a trusted root.

1. Navigate to **CAs** and find the CA in the list.
2. Click **Download Cert** to download the CA certificate as a PEM file.

Or from the **CA Detail** page, the CA certificate is displayed in PEM format and can be copied or downloaded from there.

Install this PEM file as a trusted root CA on any machine or device that needs to trust certificates signed by it.

6.4 Delete a CA

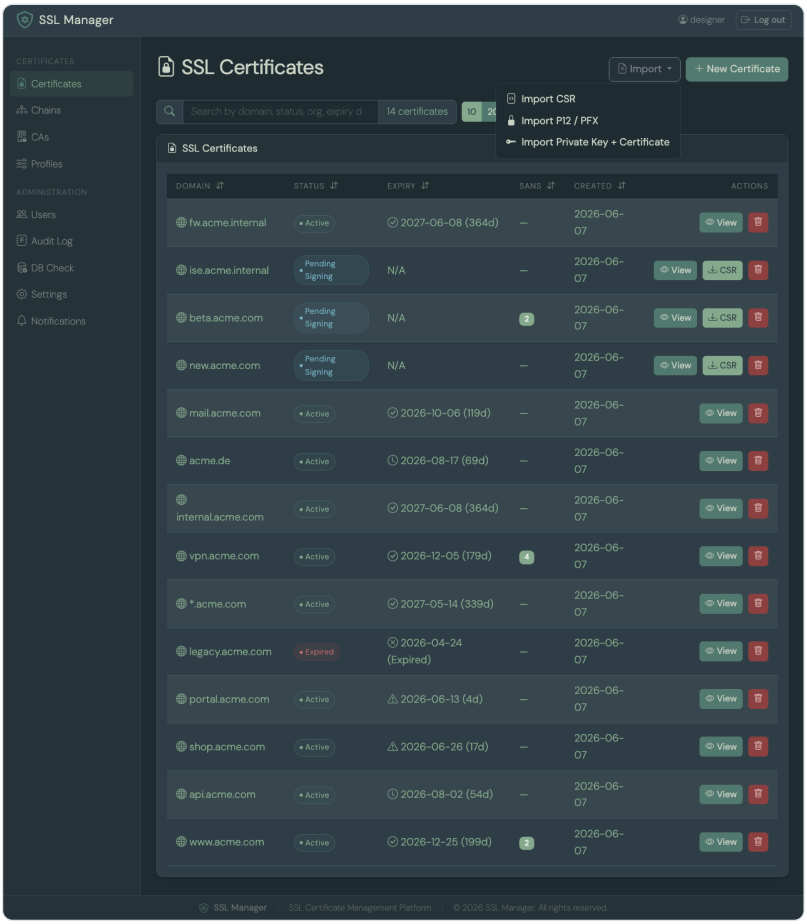
Deleting a CA removes its private key and certificate from SSL Manager. **Certificates already signed by the CA remain in the system and are not affected**, but you will no longer be able to sign new certificates with it.

1. Navigate to **CAs** and click the CA name.
2. Scroll to the **Danger Zone** section at the bottom of the CA Detail page.
3. Click **Delete CA** and confirm.

---

7. Certificate Lifecycle

The **Import** ▾ dropdown on the Certificates page groups all import paths in one place alongside the **New Certificate** button.



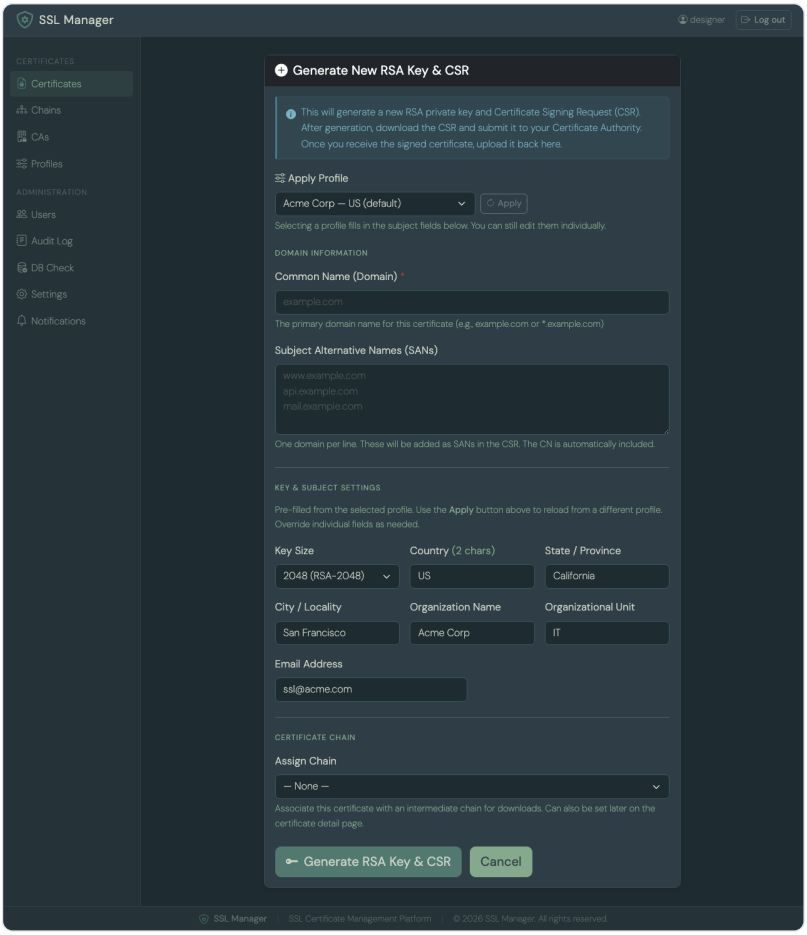
Import dropdown

7.1 Create a New Certificate

1. From the **Certificates** page, click **New Certificate**.
2. If you have multiple profiles, a **profile dropdown** appears at the top. Select the profile that matches this certificate's organisation and click **Apply** to fill the subject fields. You can edit any field after applying.
3. Fill in the form:

Field	Description
Common Name (Domain)	The primary domain this certificate covers, e.g. `www.example.com` or `*.example.com`
Subject Alternative Names	Additional domains — one per line. The CN is included automatically. Leave blank if only one domain is needed.
Key Size	Pre-filled from the profile. Change only if you need a different size for this certificate.
Country / State / City / Org / OU / Email	Pre-filled from the profile. Edit as needed for this specific certificate.
Assign Chain	Select the certificate chain that will be used when building download bundles. Can be changed later.

1. Click **Generate RSA Key & CSR**.



New certificate form

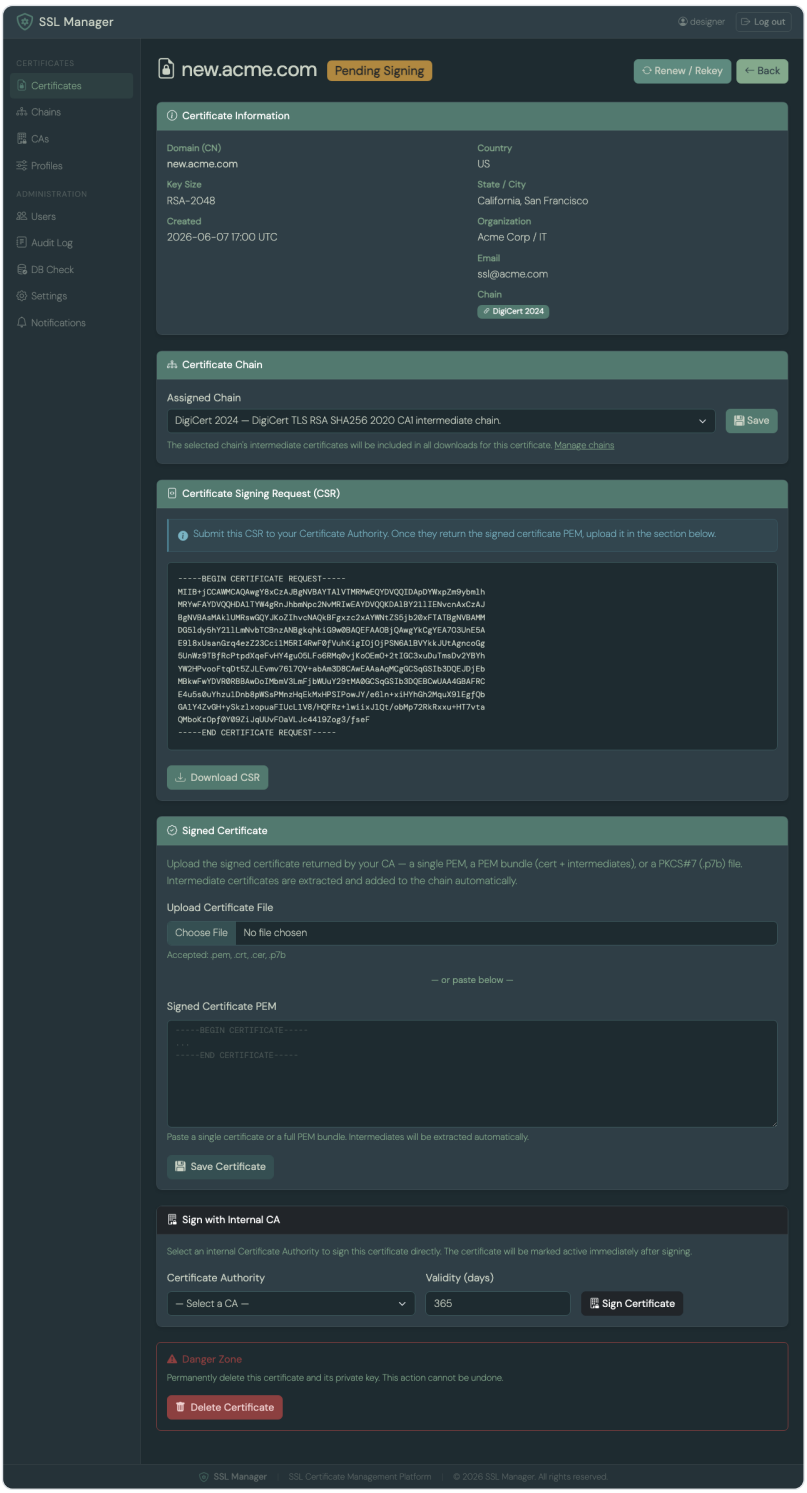
The application generates a new RSA private key (which never leaves the server) and a Certificate Signing Request. You are taken to the **Certificate Detail** page. The status shows **Pending Signing**.

7.2 Download and Submit the CSR

The CSR is what you send to your Certificate Authority (CA) to request a signed certificate.

1. On the **Certificate Detail** page, click **Download CSR**.
2. Log in to your CA's portal (e.g. DigiCert, Sectigo, GoDaddy, Let's Encrypt with a manual workflow), **or** sign the certificate using an internal CA (see Section 6.2).
3. Start a new certificate order and paste or upload the `.csr` file when the CA asks for it.
4. Complete any domain validation (DCV) steps required by the CA (email, DNS record, or file upload).
5. Once the CA approves and issues the certificate, download it. You can download either the **domain certificate only** or a **bundle that includes**

intermediates — SSL Manager handles both. The file is usually a `.crt`, `.pem`, or `.ca-bundle` file.

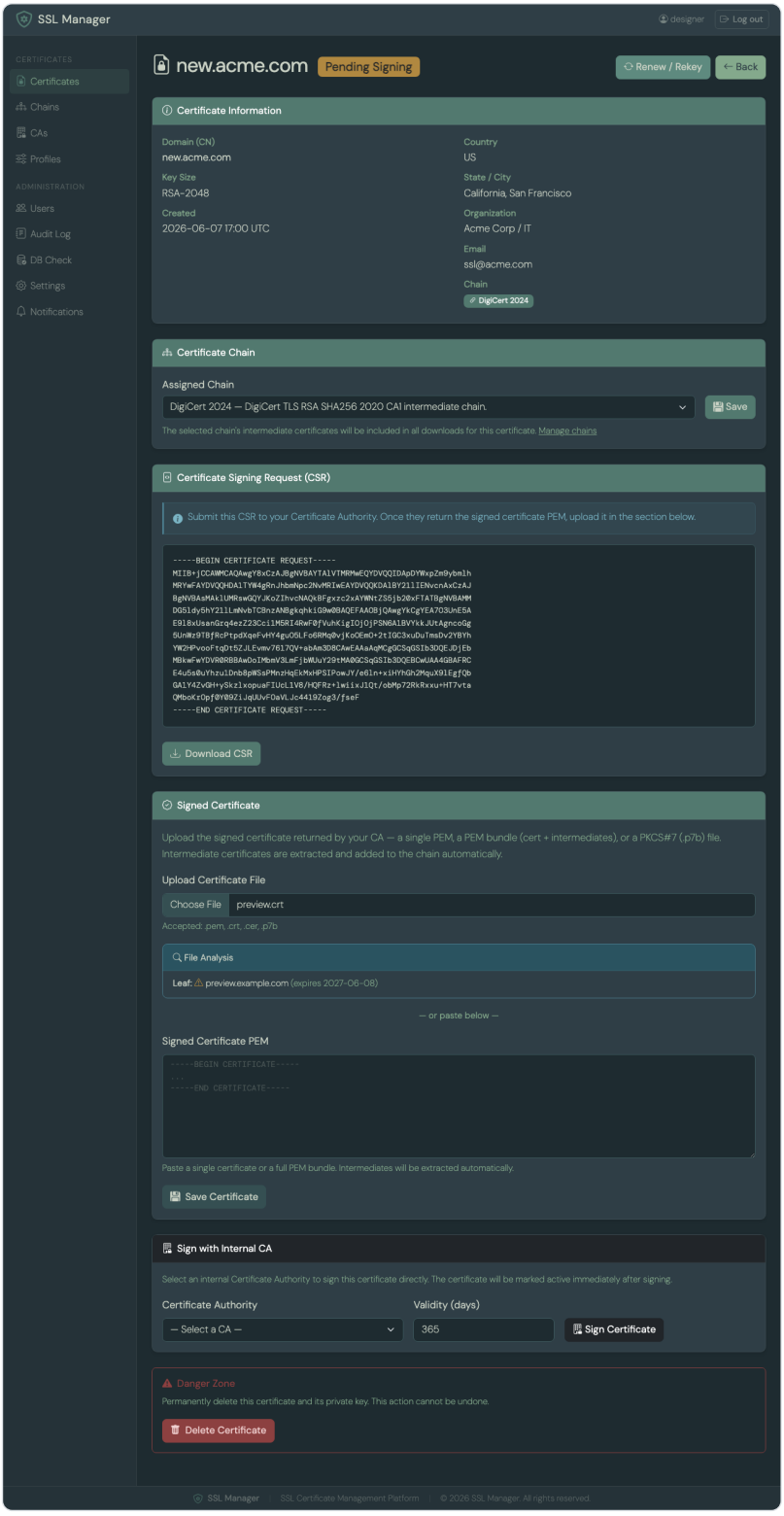


Certificate detail — pending signing, showing CSR

> **Keep the original certificate record open.** The private key is stored server-side and is already paired with this CSR. You must upload the signed certificate back to the same certificate record.

7.3 Upload the Signed Certificate

1. Return to the **Certificate Detail** page for your certificate.
2. In the **Signed Certificate** section, click **Choose File** and select the file downloaded from your CA. Accepted formats:
   - A single certificate (`.crt` , `.pem` , `.cer`)
   - A PEM bundle containing the certificate and intermediate CAs concatenated together
   - A PKCS#7 bundle (`.p7b` / `.p7`) — common from GlobalSign and some enterprise CAs
3. A **File Analysis** preview panel appears automatically, showing what was detected — the leaf certificate, any intermediates, and how the chain will be handled — before you commit.



Upload section with File Analysis preview

1. Click **Save Certificate**.

The application validates the certificate, extracts the expiry date, and stores the leaf certificate. The status changes to **Active**.

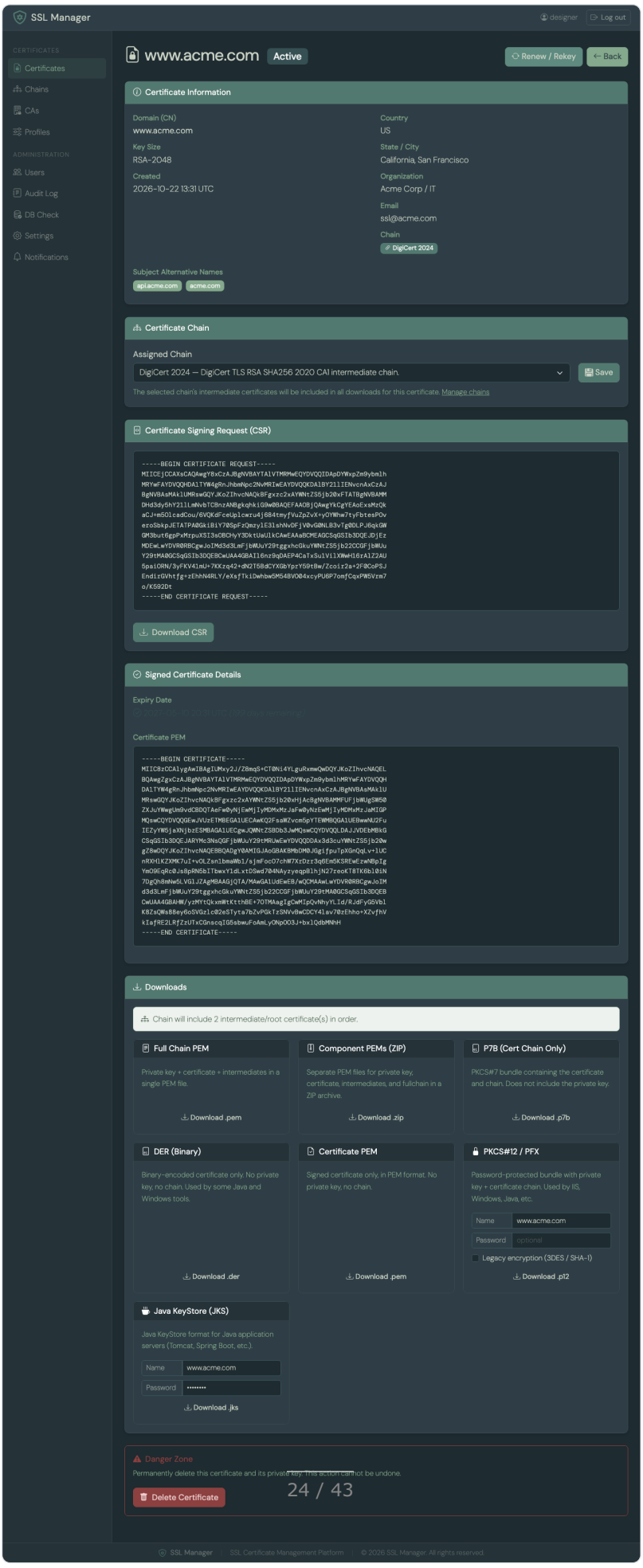
> **Bundle and P7B uploads:** If the file from your CA contains multiple certificates (common from NameCheap, Sectigo, DigiCert, GlobalSign), SSL Manager automatically identifies the leaf certificate using the certificate's CA flag and splits out any intermediates. Intermediates are added to the certificate's assigned chain, deduplicated by serial number. If no chain is assigned yet, one named `"{domain} (imported)"` is created automatically. The success message reports how many intermediates were added.

> **Key mismatch warning:** If the uploaded certificate's public key does not match the stored CSR (for example if your CA re-keyed the certificate), SSL Manager stores the certificate and shows a warning. Verify the certificate is correct before deploying it.

Colour	Meaning
Green	More than 90 days remaining
Yellow	30–90 days remaining
Red	Fewer than 30 days remaining or already expired

All download formats are now available.

SSL Manager — User Guide



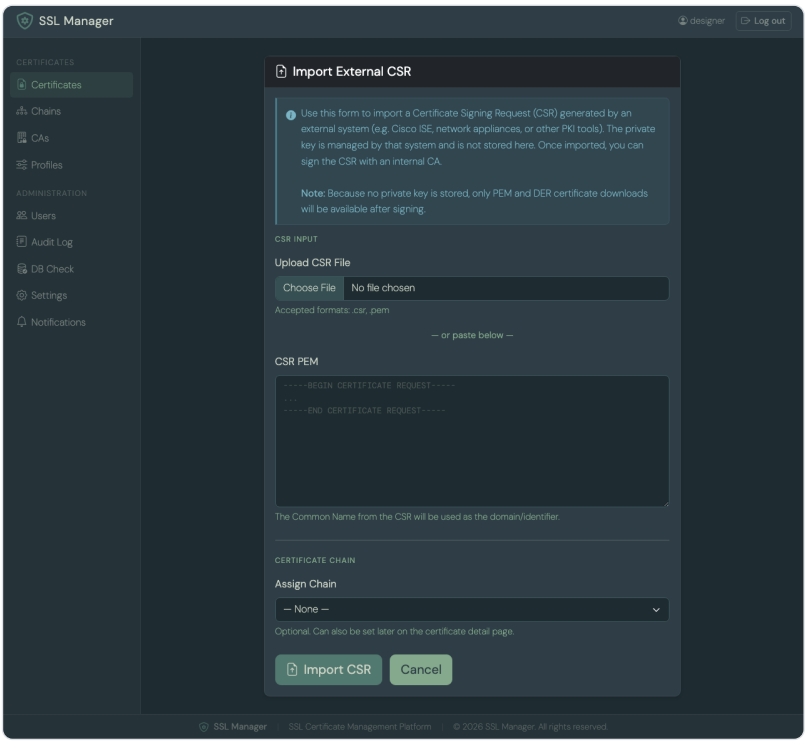
Certificate detail — active, showing all download formats

7.4 Import an External CSR

Some systems — network appliances, load balancers, identity platforms such as Cisco ISE — generate their own private key and CSR internally and export only the CSR for signing. In this case you do not have access to the private key, but you still need a signed certificate returned to the device.

Use **Import CSR** for this workflow:

1. On the **Certificates** page, open the **Import** ▾ dropdown and click **Import CSR**.
2. Either paste the CSR PEM block (including the `-----BEGIN CERTIFICATE REQUEST-----` header and footer) into the text area, or upload the `.csr` file directly.
3. Optionally assign a certificate chain.
4. Click **Import CSR**.



Import CSR form

SSL Manager parses the CSR, extracts the Common Name and Subject Alternative Names, and creates a certificate record with status **Pending Signing**. The private key is **not** stored — it remains on the originating device.

Signing an imported CSR:

The imported certificate can be signed in the same way as any other pending certificate: - Use an internal CA (see Section 6.2). - Or download the CSR and submit it to an external CA, then upload the signed certificate back.

Download restrictions for imported CSRs:

Because the private key is not held by SSL Manager, formats that require it are unavailable:

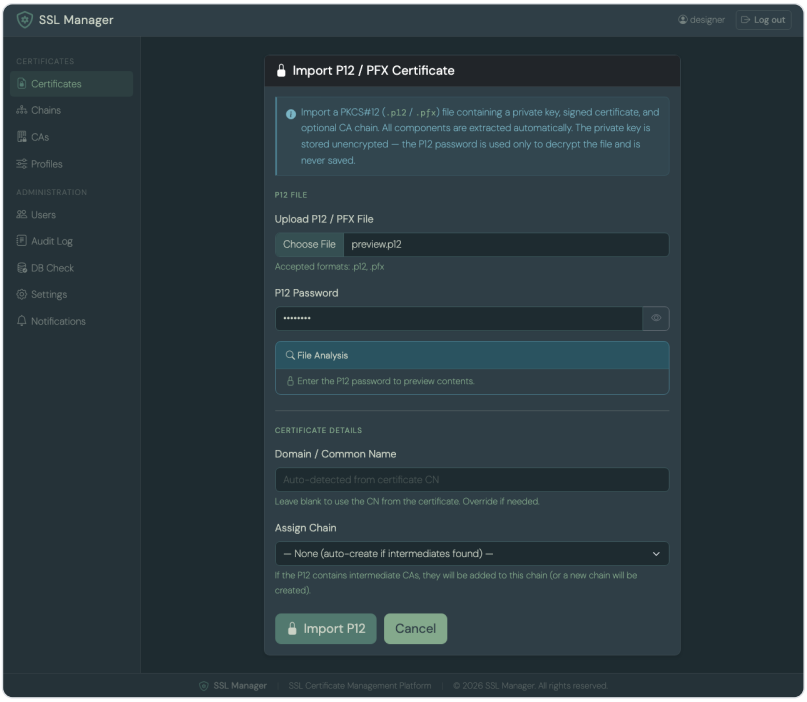
Format	Available?
Certificate PEM	Yes
DER	Yes
Full Chain PEM	No (requires private key)
Component ZIP	No (requires private key)
PKCS#12 / PFX	No (requires private key)
Java KeyStore (JKS)	No (requires private key)
P7B	No (requires private key)

Download the **Certificate PEM** and install it on the originating device following that device's certificate import procedure.

7.5 Import a PKCS#12 / PFX Certificate

Use this when you already have a certificate and its private key bundled together in a `.p12` or `.pfx` file — for example, when migrating from another certificate management tool or exporting from a Windows certificate store.

1. On the **Certificates** page, open the **Import** ▾ dropdown and click **Import P12 / PFX**.
2. Click **Choose File** and select your `.p12` or `.pfx` file.
3. Enter the **P12 Password**. As you type, a **File Analysis** preview appears showing the private key type, the certificate details, any intermediates, and how the chain will be handled.



Import P12 form with File Analysis preview

1. Optionally override the **Domain / Common Name** (leave blank to use the CN from the certificate).
2. Optionally select a chain to assign.
3. Click **Import P12**.

A new certificate record is created with status **Active**. The private key is extracted from the P12, decrypted, and stored unencrypted by SSL Manager — the P12 password is used only during import and is never saved.

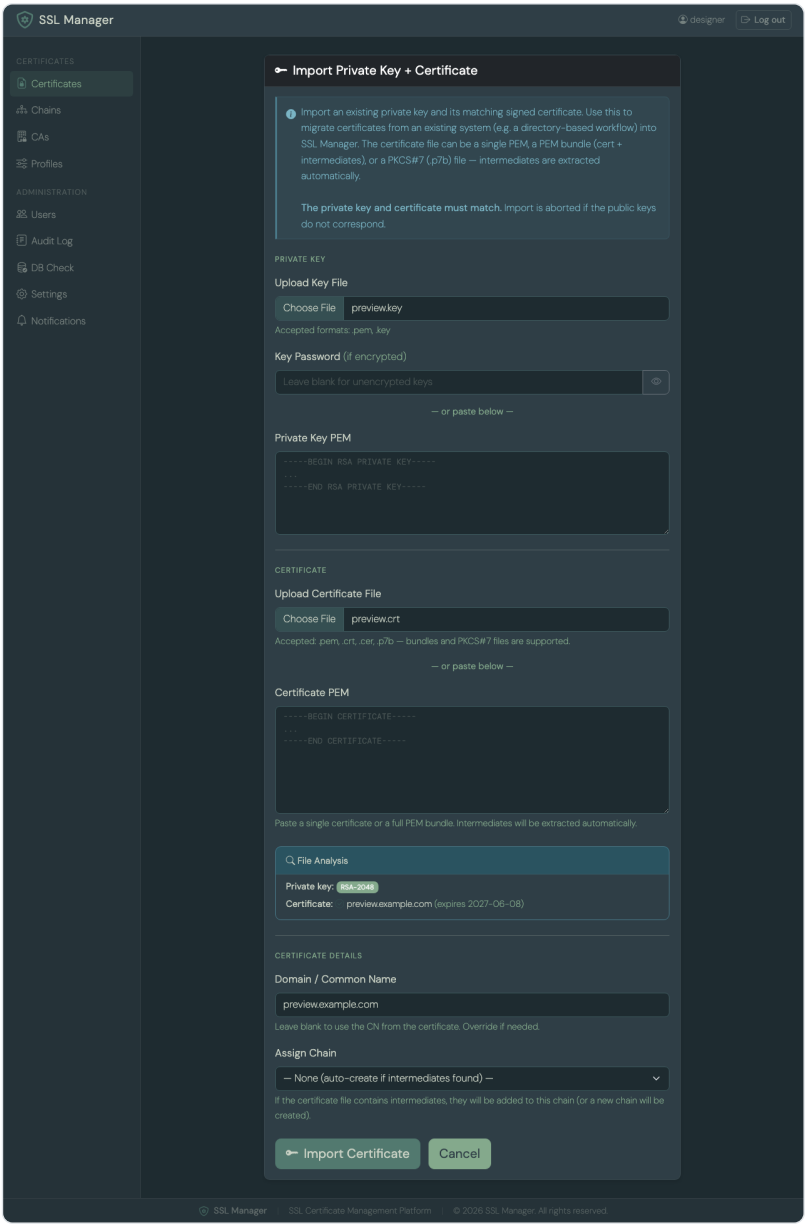
> **Chain matching:** If the P12 contains intermediate CA certificates that match an existing chain in SSL Manager, that chain is assigned automatically. If no match is found, a new chain named `"{domain} (imported)"` is created.

7.6 Import a Private Key + Certificate

Use this when you have a private key and a signed certificate as separate PEM files — common when migrating from a directory-based certificate workflow or another tool that stores them separately.

1. On the **Certificates** page, open the **Import** ▾ dropdown and click **Import Private Key + Certificate**.
2. Upload the **private key file** (`.pem` or `.key`), or paste the key PEM into the text area. If the key is encrypted, enter the key password.

3. Upload the **certificate file**, or paste the PEM into the text area. This can be a single certificate, a PEM bundle with intermediates, or a `.p7b` file.
4. Once both are provided, a **File Analysis** preview appears confirming the key and certificate match, showing any detected intermediates, and describing the chain action.



Import keypair form with File Analysis preview

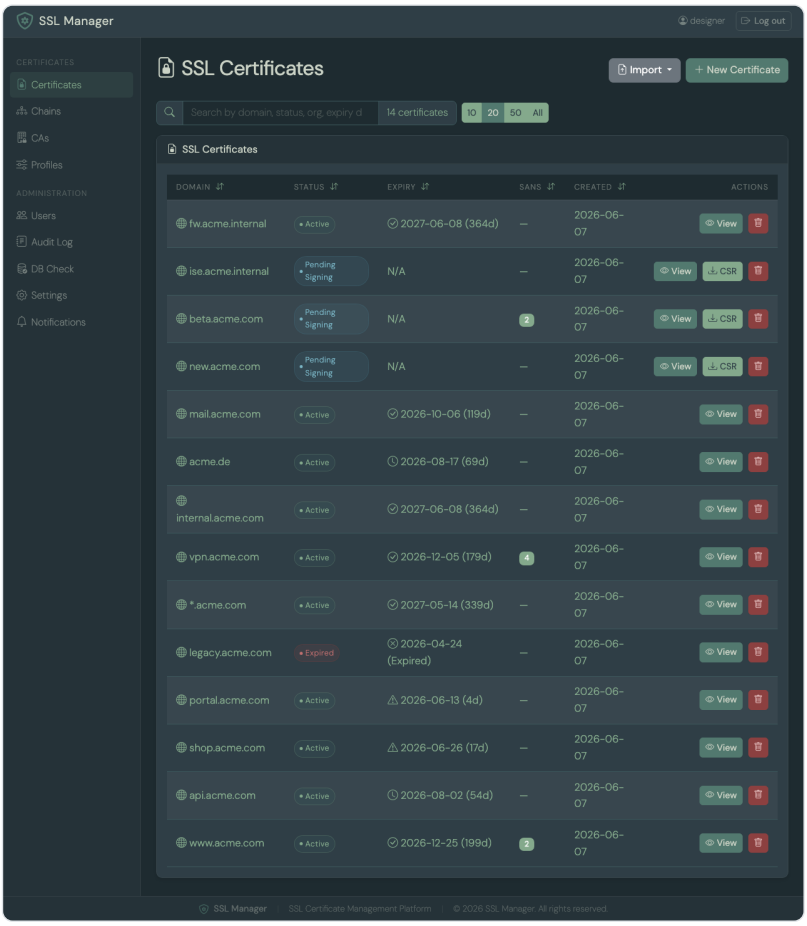
1. Optionally override the domain and assign a chain.
2. Click **Import Certificate**.

> **Key/certificate mismatch:** If the private key and certificate do not correspond (their public keys differ), the import is blocked with an error. Verify you have the correct key and certificate pair before trying again.

---

8. Browsing and Finding Certificates

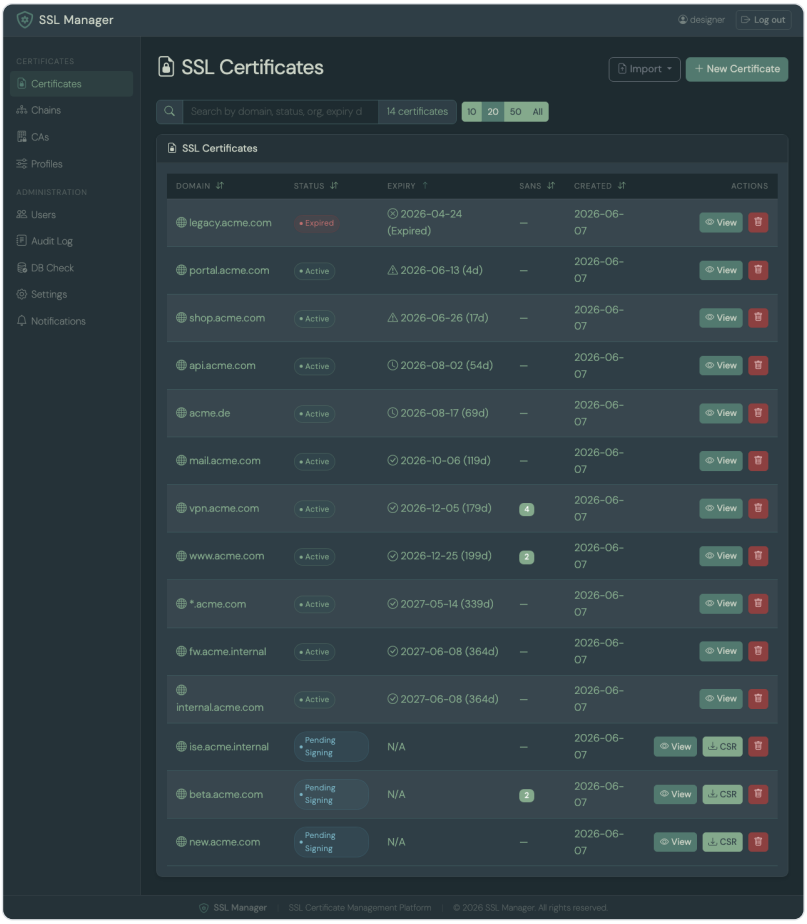
The **Certificates** page lists all certificates with their domain, status, expiry, SANs count, and creation date.



Certificates list — default view

Sorting

Click any column header to sort by that column. Click again to reverse the sort direction. An arrow icon indicates the active sort column. The default sort is **Created** (newest first).



Certificates list sorted by expiry — oldest first

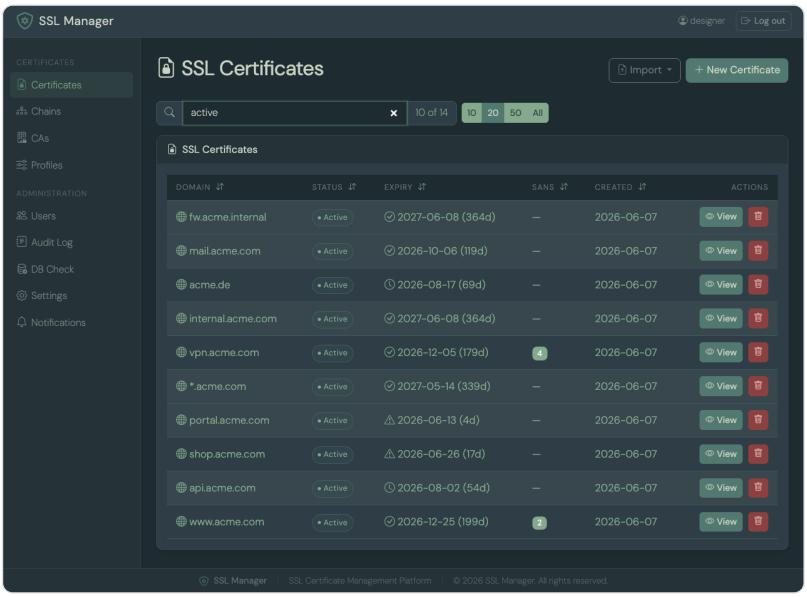
Searching

Type in the **search bar** above the table to filter certificates in real time. The search checks:

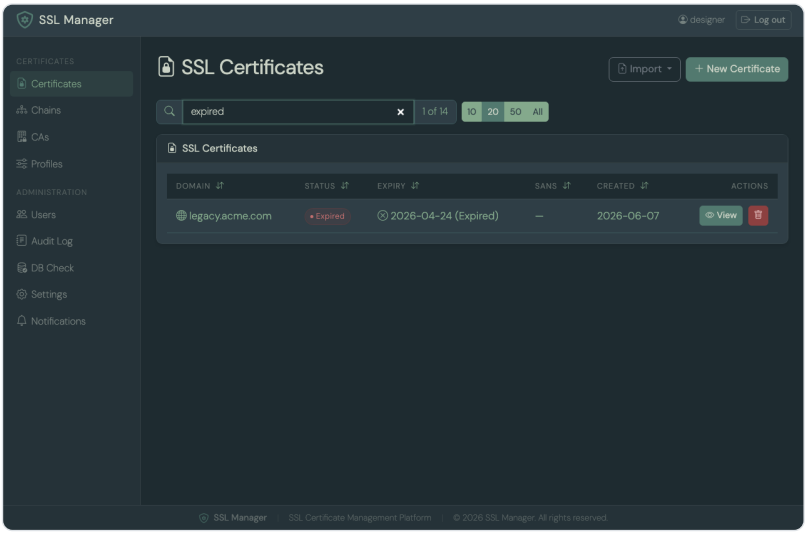
- Domain name
- Status (`active` , `expired` , `pending signing`)
- Organisation name
- Country code
- Email address
- Expiry date (in `YYYY-MM-DD` format)
- Creation date
- SAN domains

The counter on the right of the search bar shows how many certificates match (e.g. `3 of 12 certificates`). Clear the search box to show all certificates again.

Examples: - Type `expired` to see all expired certificates at a glance. - Type `example.com` to find all certificates covering that domain. - Type `2025-` to list certificates expiring or created in 2025. - Type `acme` to find all certificates tied to that organisation.



Search filtering to show only active certificates



Search filtering to show only expired certificates

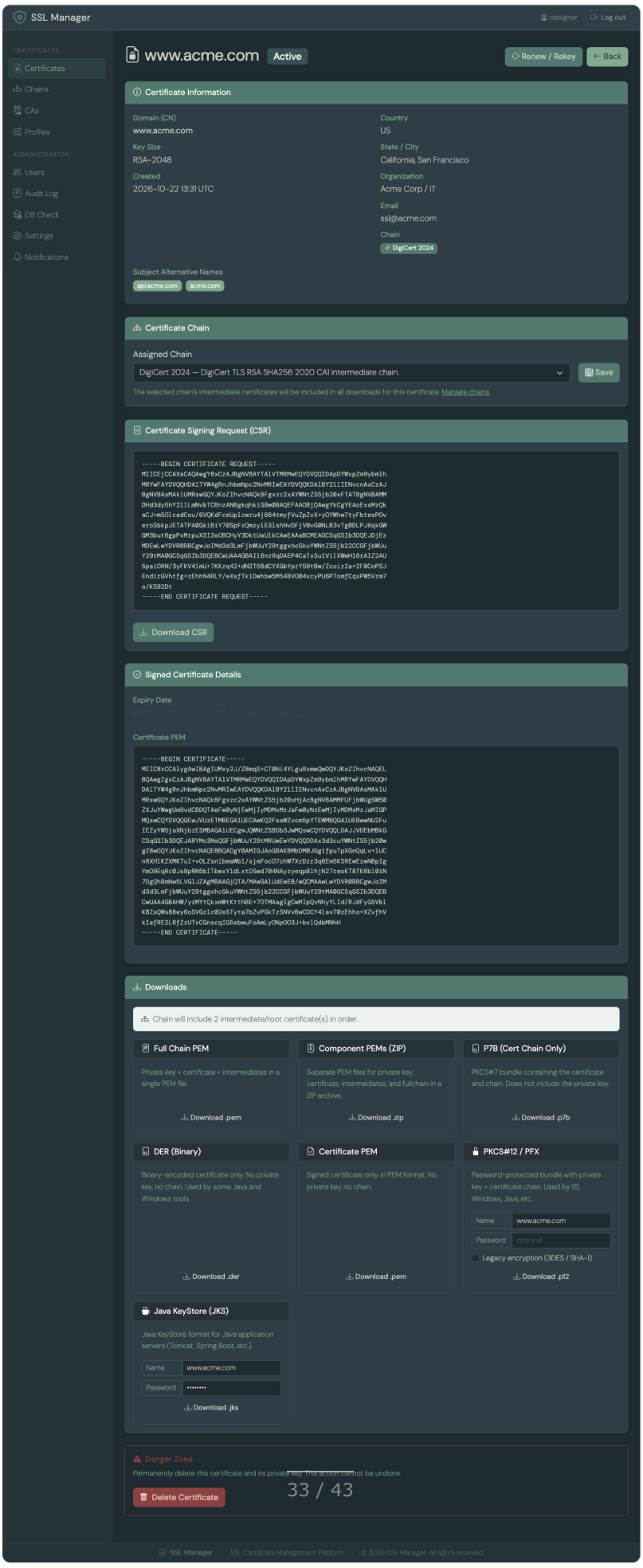
Rows per page

Use the **10 / 20 / 50 / All** button group (top-right of the toolbar) to control how many certificates are shown at once. The default is 20. When more rows exist than the selected page size, a page navigation bar appears below the table.

9. Downloading Certificates

Once a signed certificate has been uploaded, the **Downloads** section on the **Certificate Detail** page offers several formats. Choose the format that matches your server or platform.

SSL Manager — User Guide



Certificate detail — downloads section

Full Chain PEM

Button: Download .pem **File:** `domain-fullchain.pem`

Contains the private key, signed certificate, and all intermediate certificates in a single file. Use this for **HAProxy**, some CDN origin configurations, and any tool that requires everything in one PEM file.

Component ZIP

Button: Download .zip **File:** `domain-certs.zip`

A ZIP archive containing individual files for maximum flexibility:

File	Use
`private_key.pem`	nginx `ssl_certificate_key`, Apache `SSLCertificateKeyFile`
`certificate.pem`	The domain certificate on its own
`chain.pem`	Intermediate certificates only — Apache `SSLCACertificateFile`
`fullchain.pem`	Domain certificate + intermediates (no key) — nginx `ssl_certificate`, Apache `SSLCertificateFile`

PKCS#12 / PFX

Button: Download .p12 **File:** `domain.p12`

A password-protected bundle containing the private key, certificate, and chain in a single encrypted file.

1. Enter a password in the **Password** field before clicking Download.
2. Store the password securely — it is required every time the `.p12` file is imported.

Use for: **Windows IIS**, Azure App Service, F5 BIG-IP, and any platform that imports a PFX file.

Legacy encryption (3DES / SHA-1): off by default. By default the bundle is encrypted with AES-256 and a SHA-256 MAC, which **Windows Server 2012 R2 and 2016** cannot import (the import fails with a password error even when the password is correct). Tick this box to produce a bundle encrypted with PBE-

SHA1-3DES and a SHA-1 MAC instead, the same as `openssl pkcs12 -export -certpbe PBE-SHA1-3DES -keypbe PBE-SHA1-3DES -macalg sha1`. Use it only for systems that need it; the default is stronger.

Java KeyStore (JKS)

Button: Download .jks **Fields:** Store password (default: `changeit`), Alias (default: `certificate`) **File:** `domain.jks`

Use for: **Tomcat**, Spring Boot, JBoss/WildFly, Jetty, and any Java application server.

P7B (Chain Bundle, No Private Key)

Button: Download .p7b **File:** `domain.p7b`

Contains the signed certificate and all intermediate certificates. Does **not** include the private key.

Use for: **Windows Server** (MMC certificate store), IIS when the private key is already present, some enterprise CA portals.

DER (Binary Certificate)

Button: Download .der **File:** `domain.der`

The domain certificate in binary DER encoding. Does not include the private key or chain.

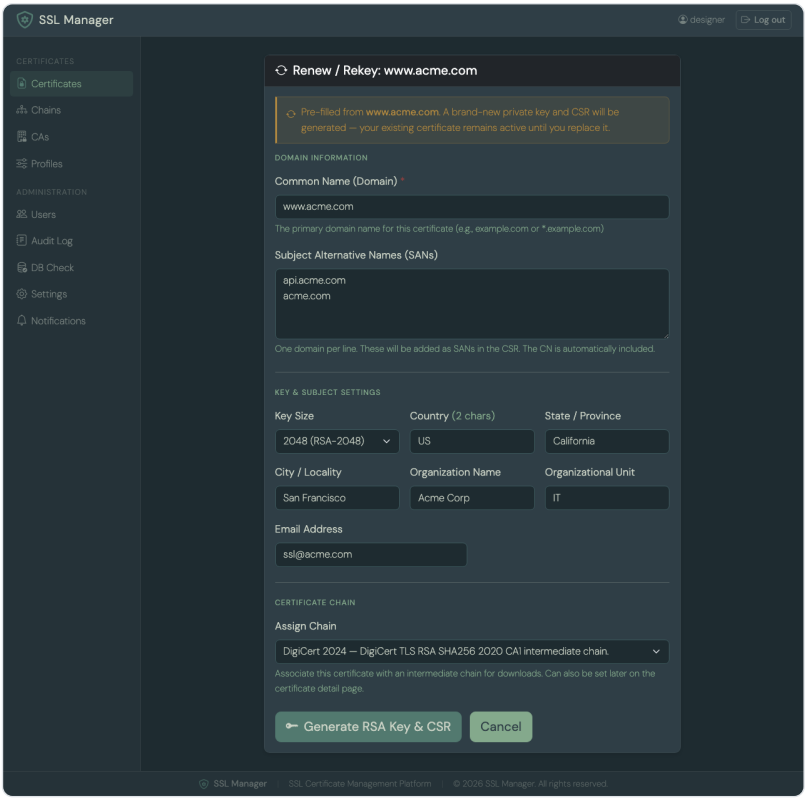
Use for: Java `keytool` imports, some embedded devices, and systems that require a binary certificate file.

---

10. Renewing a Certificate

Certificates typically need to be renewed annually. SSL Manager's renew/rekey workflow generates a fresh private key and CSR while keeping your existing certificate active until you are ready to replace it.

1. On the **Certificate Detail** page, click **Renew / Rekey**.



Renew / Rekey form — pre-filled from the existing certificate

1. The form pre-fills with the existing domain, SANs, subject fields, and chain assignment. Edit any field that has changed (e.g. updated organisation name or a new SAN).
2. Click **Generate RSA Key & CSR**.

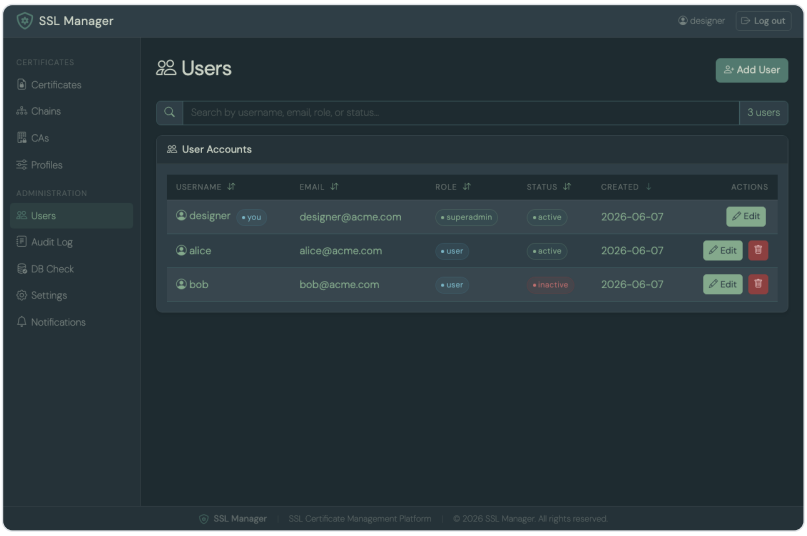
A new certificate record is created with status **Pending Signing**. Your original certificate remains **Active** and continues to serve traffic.

1. Follow the same steps as a new certificate: download the CSR, submit it to your CA, upload the returned signed certificate.
2. Once the new certificate is verified and deployed, return to the **Certificates** page and delete the old record.

> **Tip:** Start renewals at least 30 days before expiry. The expiry badge changes from green to yellow at 90 days and red at 30 days as a visual reminder.

11. User Management

User management is available to **superadmin** accounts only. Navigate to **Users** in the navbar.



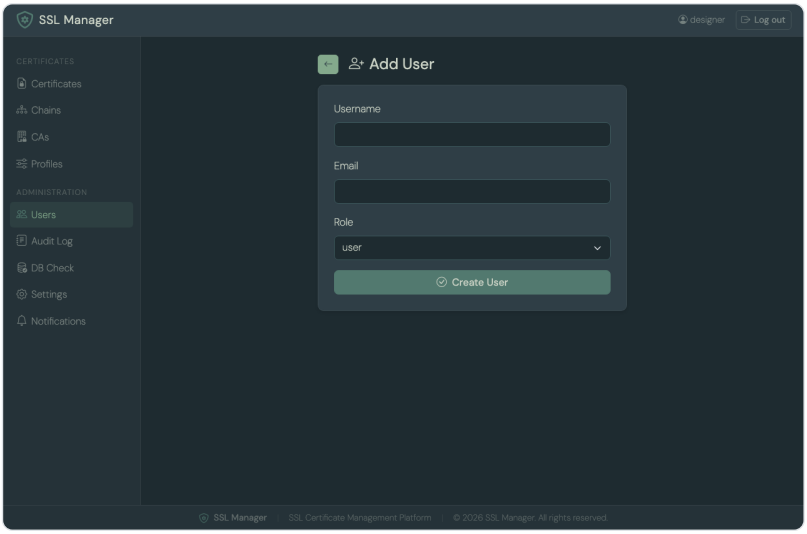
Users list

Roles

Role	Capabilities
`superadmin`	Everything a user can do, plus manage users and view the audit log
`user`	Create and manage certificates, chains, and profiles; download all formats

Add a User

1. Click **Add User**.
2. Fill in username, email, password, confirm password, and role.
3. Click **Create User**.

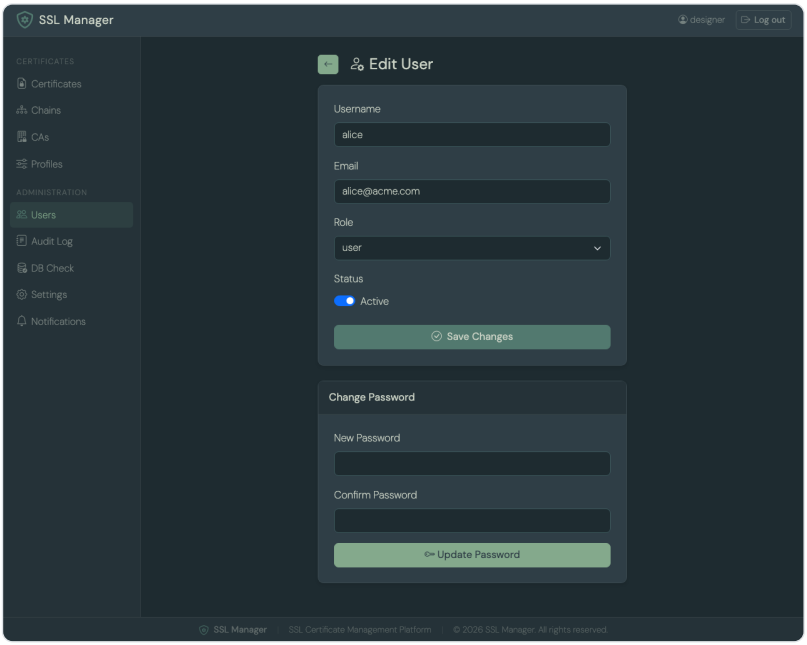


Add user form

Edit a User

Click the pencil icon next to a user. You can update: - Username and email - Role (`superadmin` or `user`) - Active status — deactivating a user prevents login without deleting the account

To change a password, fill in the **New Password** and **Confirm Password** fields. Leave them blank to keep the current password.



Edit user form

Delete a User

Click the trash icon next to a user. You cannot delete your own account.

Superadmin Protection

At least one active superadmin must always exist. The following are blocked when only one active superadmin remains: - Deleting the last superadmin - Changing the last superadmin's role to `user` - Deactivating the last superadmin's account

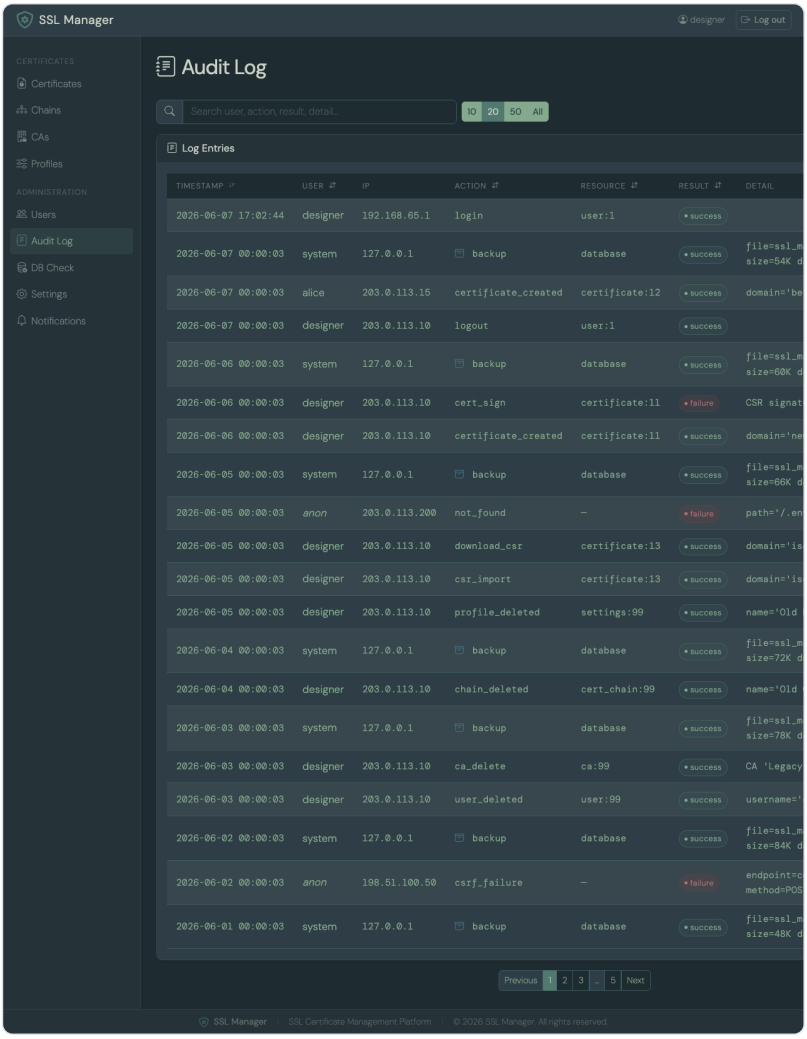
To transfer superadmin responsibilities, promote a second user to `superadmin` first, then make changes to the original account.

---

12. Audit Log

The **Audit Log** is available to **superadmin** accounts only. Navigate to **Audit Log** in the navbar.

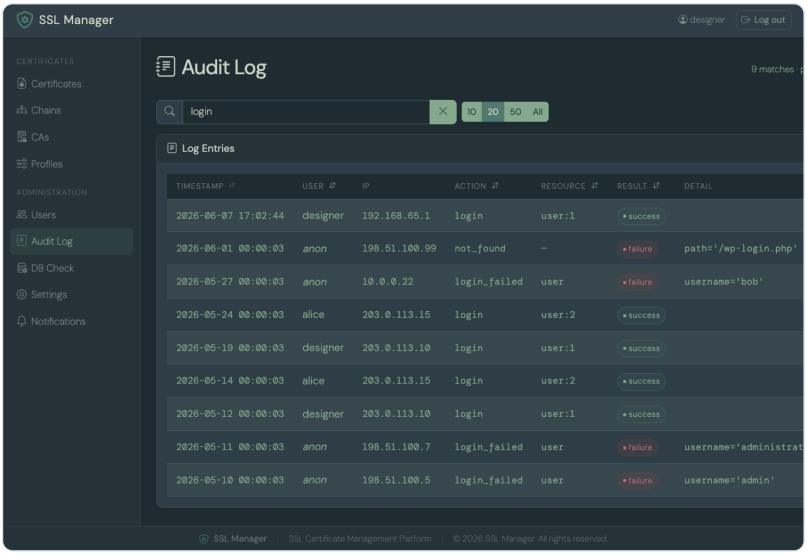
Every action taken in the application is recorded — logins, certificate creation, signing, downloads, user changes, and automated system events such as backups.



Audit log

Searching

Type in the search bar to filter log entries in real time across all columns (user, action, resource, result, and detail). Click the × button or clear the search box to reset.



Audit log filtered by search term

Sorting

Click any underlined column header to sort by that column. Click again to reverse the direction.

Rows per page

Use the **10 / 20 / 50 / All** button group to control how many entries are shown per page. The default is 20. Page navigation appears below the table when there are more entries than the current page size.

Result badges

Badge	Meaning
`success`	The action completed without errors
`failure`	The action failed (detail column gives the reason)
Other	Informational status specific to the action type

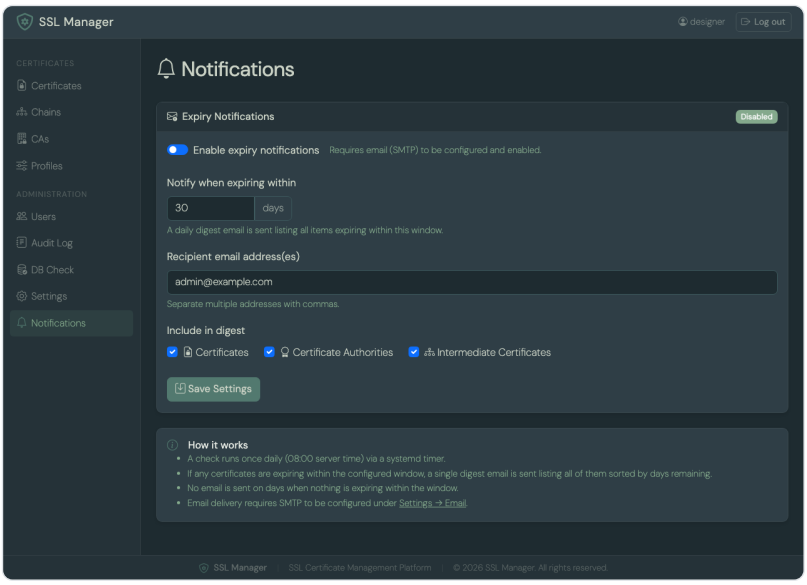
---

13. Expiry Notifications

Expiry notifications are available to **superadmin** accounts only. Navigate to **Notifications** in the navbar.

When enabled, SSL Manager sends a daily digest email listing all certificates, CAs, and intermediates expiring within a configurable window. The email is only sent on days when at least one item is found — no email is sent when nothing is expiring within the threshold.

> **Prerequisite:** Email must be configured and enabled under **Settings → Email** before notifications can be delivered. See your administrator if SMTP is not yet set up.



Notifications settings page

Configuration

Field	Description
Enable expiry notifications	Master switch. When off, no emails are sent regardless of other settings.
Notify when expiring within	How many days before expiry to start including an item in the digest. Default is 30 days.
Recipient email address(es)	One or more email addresses to send the digest to. Separate multiple addresses with commas.
Include in digest	Choose which certificate types to include: leaf certificates, internal CAs, and/or intermediate certificates.

The digest email

The digest lists all matching items sorted by days remaining, with colour-coded badges:

Badge colour	Meaning
Red `EXPIRED`	Certificate has already passed its expiry date
Red `Nd`	Expires in 7 days or fewer
Orange `Nd`	Expires in 8–14 days
Yellow `Nd`	Expires in 15 days or more (within threshold)

Each notification send (success or failure) is recorded in the **Audit Log** under the action `expiry_notification_sent`.

For installation, deployment, and system administration see README.md. For detailed CLI verification commands and CA-specific notes see WORKFLOW.md.